

COMETS: Constellation of Meteorological Satellites for global weather coverage

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The Integrated Team Project (ITP) required building a constellation of satellites that could perform meteorological measurements using GNSS Radio Occultation (GNSS-RO) and Microwave sounder (MWS) sensors. The satellites were built with a shared platform called COCON to reduce costs and simplify production.

Weather Satellite Constellation | GNSS-RO | Microwave Sounder

Weather data is important for climate research, as it helps scientists understand atmospheric and climatic processes and predict short- and long-term weather conditions. It also has implications for human activities such as agriculture, transportation, and energy. Weather observation satellites are useful for tracking hazardous weather events like hurricanes and typhoons, allowing authorities to protect people and property. COMETS is a project focused on a large constellation of weather observation satellites with two types of sensors: GNSS-RO and MWS, which offer a better spatial and temporal resolution of weather data and reliability.

1. Mission

To achieve global Earth coverage in 20 minutes or less, a Walker constellation configuration was used with five planes and eight satellites per plane (see Figure 1). Sun-synchronous orbits (SSOs) were chosen for optimal illumination conditions for the solar array and to simplify thermal design for thermal control. An orbit altitude of 650 km was chosen to meet requirements while minimizing drag perturbations and the need for lifetime extension maneuvers. This altitude allowed for an inclination of $i \approx 98^\circ$, which could be achieved by a launcher like the Ariane 6.

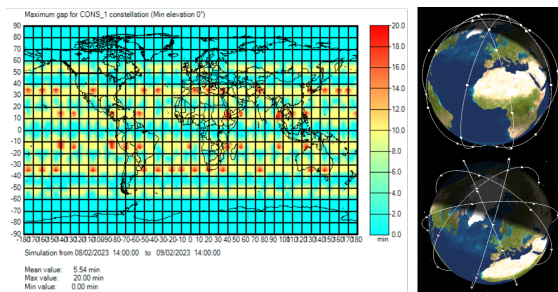


Fig. 1. Walker constellation configuration of 40/5/1

The GNSS-RO sounding method uses radio signals from GNSS satellites to determine the Earth's atmosphere temperature and water vapor profiles. The LION 1300Neo, manufactured by Airbus, is the instrument onboard our satellites that processes the data and calculates the weather data. The LION 1300Neo is small and has a 15-year lifetime, making it ideal for the mission's 10-year duration. It can receive multi-constellation signals, including GALILEO, GPS, Glonass, and Compass. Microwave sounding, which measures the thermal emission of the atmosphere at different frequencies to obtain information about the atmosphere's vertical temperature and moisture structure, is achieved through the BOWIE-M MW sounder from Ball Aerospace, a technology with a TRL 6, integrated into our satellites.

2. COCON: Interactions between constellations

The COCON initiative was an idea of the Master TAS ASTRO students in order to work on common features and have interactions between projects. Main advantages are: More people involved in each subsystem gives different insight and ideas to give refined and more robust results and building a common platform means a unique production thus lower the global cost. The common structure hexagonal shape was mainly driven by constellation using optical instruments with high pointing accuracy need. Materials used are aluminium for the primary structure and carbon/epoxy composite for the appendages and tanks.

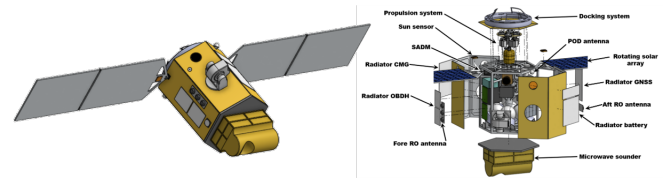


Fig. 2. Shared Platform

To meet the high thrust requirements of the optical constellation, four thrusters will be used. Exotrail's micro-XL modular solution allows for easy addition of more thrusters, improving propulsion system reliability and reducing maneuver time. The COCON initiative's Mover constellation, designed by another ITP group, offers flexibility and will be used for COMETS missions:

- Early Orbit Phase: Phasing without the use of our thrusters.
- End Of Life: Perform the deorbitation of the COMETS constellation without the use of our own thrusters.
- Allows us flexibility by doing RAAN manoeuvres to extend the lifetime of the constellation in case of failure.

Europe will be able to access its own meteorological data globally and achieve sovereignty in this field through the use of COMETS. This constellation is rare for meteorological applications and was made possible through the COCON global initiative.

3. COMETS technical features

Orbit	Circular SSO, Alt = 650 km, $i \approx 98^\circ$
Instruments	MWS: BOWIE-M (Ball Aerospace) GNSS-RO: LION 1000 SERIES (Airbus)
Mass	<350 kg
Architecture	Hexagonal
Lifetime	> 10 years
Avionics	Spaceteq OBC-GR712 (LEO3FT processor)
Communication	LASER: Condor MK3
Thrusters	Exotrail micro-XL
Battery	SAFT VES16 Li-ion cell
Solar Array	3G30C cell (Azure Space)

Table 1. COMETS Satellites key features